

Consumption of *Syzygium gratum* Promotes the Antioxidant Defense System in Mice

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Abstract Several vegetables have been shown to possess cytoprotective and antioxidant effects with various mechanisms of action. The aim of this study was to determine the antioxidant effects and mechanism underlying of *Syzygium gratum*, a dietary and herbal plant commonly found in the Southeast Asia. Additionally, its effects on the induction of endogenous antioxidant defensive system were also investigated. Results showed that the leaf extract possessed an exceptionally strong antioxidant and intracellular oxygen radical scavenging activity in both aqueous and ethanolic extracts. The plant aqueous extract was further studied in C57BL/6J mice to evaluate its effects *in vivo*. The extract was well tolerated by the animals throughout the 30 days of study. The cytoprotective enzyme, heme oxygenase (HO-1) activity was significantly increased in the high dose-treated animals (1 g/kg/day). Consistent with the enzymatic activity, the expression of HO-1 mRNA tended to increase in those mice. There was no significant increase in hepatic γ -glutamylcysteine ligase (γ -GCL) activity, glutathione levels and GCL mRNA expression. Taken together, this study provides evidence that *S. gratum* exhibits potent direct antioxidant properties and can induce cytoprotective enzyme *in vivo*. Consumption of *S. gratum* may provide a health benefit against oxidative stress and other related disorders.

Keywords Antioxidant · Cytoprotection · γ -glutamylcysteine ligase · Heme oxygenase · Natural product · *Syzygium gratum*

Abbreviations

DHF	dihydrofluorescein diacetate
DPPH	1,1-diphenyl-2-picrylhydrazyl
FRAP	ferric reducing antioxidant power
γ -GCL	γ -glutamylcysteine ligase
GSH	glutathione
HO-1	heme oxygenase-1

Introduction

An imbalance between the generation of oxygen derived reactive species and the levels of antioxidant defense enzymes or compounds may lead to the condition called oxidative stress. This condition elicits cellular toxicity involving various targets such as DNA, proteins and lipids. In man, reactive oxygen species-induced cellular damage has been implicated in the development of the ageing process, several chronic diseases involving with inflammation, cardiovascular diseases and cancers [1]. Cumulative evidences in experimental animals and clinical studies have shown that supplementation of antioxidants could be useful in prevention and management of oxidative stress related disorders [2, 3].

Fruits and vegetables contain several hundred different types of antioxidants that may directly react with toxic oxygen or nitrogen species forming products with much lower reactivity [4, 5]. Alternatively, compounds in a plant-based diet may increase the capacity of the endogenous antioxidant defenses by modulating genes related to

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antioxidant and cytoprotective enzymes [6, 7]. A number of studies have demonstrated that the expression of protective enzymes such as heme oxygenase (HO-1) and γ -glutamylcysteine ligase (γ -GCL) could be induced by dietary antioxidants via activation of redox-sensitive transcriptional factors [8–10].

HO-1 is an enzyme responsible for heme metabolism and plays critical roles as a cytoprotective and anti-inflammatory enzyme [11]. It has been implicated as a protective mediator in numerous experimental models of disease states and in clinical medicine including sepsis and ischemic reperfusion injuries. Various phytochemical compounds including caffeic acid phenethyl ester, resveratrol and curcumin, and oxidant stress elicited, for instances by ischemia-reperfusion, may act as inducers of the HO-1 gene [10–12]. Similar to HO-1, γ -GCL has a potential to protect cells against damage caused by oxidative stress. The enzyme catalyzes the rate-limiting step of the synthesis of glutathione (GSH) which is the most abundant intracellular antioxidant molecule. Previous studies have demonstrated that dietary compounds such as curcumin and beta-naphthoflavone could induce the expression of γ -GCL [8, 13]. Thus, several phytochemicals may exert their cytoprotective effects by mediating the expression of these two enzymes.

Syzygium gratum (*S. gratum*, synonym: *Eugenia grata* Wight), family Myrtaceae, is a widely consumed local dietary and herbal plant in the Southeast Asian countries. Its leaves are usually consumed fresh as side dish and they have highly nutritional value of high protein, iron and beta-carotene contents. Moreover, the plant is used as health promoter for the relief of arthralgia, dyspepsia, indigestion, peptic ulcer, diarrhea, bacterial infection, asthma, and cardiovascular diseases [14, 15]. Recently, we have shown that *S. gratum* provides cardiovascular protection against phenylhydrazine-induced vascular endothelial dysfunction, by alleviation of oxidative stress and regain in the vascular responses [16]. In the present study, we have investigated the antioxidant effects of *S. gratum*. Our results demonstrate that the plant possesses an exceptionally high antioxidant effects which are associated with induction of the HO-1 enzyme.

Materials and Methods

Plant Material and Extraction

Fresh leaves of *S. gratum* were collected from local agricultural fields in and around Khon Kaen Province, Thailand. It was identified as *S. gratum* by the same method as described in our previous study [16]. To prepare the aqueous extract of *S. gratum*, the leaves of plant were weighed, chopped and boiled in purified water for 30 min,

and then were dried into a powder using a spray-drying technique which yielded a residue of 16.7% by dry weight of plant. In a preliminary study of preparation of the aqueous extract, we observed that the extract prepared by spray-drying method gave a comparable antioxidant activity as by a usual freeze-drying method (data not shown). The spray-drying method was subsequently used for preparation of the aqueous extract. For ethanolic extract, the dried leaves were extracted with 95% ethanol until exhaustion. The extract obtained was filtered and concentrated in a rotary evaporator yielding a residue of 8.9% by dry weight.

1,1-Diphenyl-2-picrylhydrazyl (DPPH) Radical Scavenging Assay

The free radical scavenging activity of the extract was assayed by DPPH according to a previously described method [16]. The test compound solutions (1.25–1,250 μ g/mL) were added to DPPH (0.1 mM) solution. The mixture was incubated for 20 min at room temperature and the absorbance at 515 nm was measured. The free radical scavenging activity of 0.1–100 μ g/ml Trolox (a water-soluble derivative of vitamin E; 6-hydroxy-2.5.7.8-tetramethylchroman-2-carboxylic acid) was also measured to compare the relative potency of the extracts.

Dihydrofluorescein (DHF) Assay

The intracellular antioxidant activity of the extracts was determined in rat peritoneal macrophages according to the method previously described [16] with some modifications. Briefly, 10^6 cells/ml of peritoneal macrophages from male Sprague–Dawley rats in Hank's buffer supplemented with 15 mM HEPES were incubated with catalase (1,000 U) for 10 min at 37 °C, followed by the plant extracts, dihydrofluorescein diacetate, and phorbol-12-myristate-13-acetate (0.6 μ M). After incubation for 45 min, fluorescent readings were made with excitation and emission wavelengths of 485 and 520 nm, respectively.

Ferric Reducing Antioxidant Power (FRAP) Assay

The FRAP assay was performed by using Benzie and Strain's method [17] with some modifications. The freshly prepared FRAP reagent comprised 300 mM acetate buffer, pH 3.6; 10 mM 2,4,6-tripyridyl-s-triazine in 40 mM HCl; and 20 mM FeCl_2 in the ratio 10:1:1. FRAP reagent was mixed with the extracts of *S. gratum* using ascorbic acid for the calibration of FRAP assay. The mixtures were incubated for 4 min at 30 °C. The change in absorbance was monitored at 610 nm, and was translated into a FRAP value. Antioxidant activity was calculated and expressed as ascorbic acid equivalent per mg of extract (μ g/mg extract).

Total Phenolics Assay

The assay is based on a reduction of Folin-Ciocalteu reagent and samples were calibrated against gallic acid as a standard (0.05–0.5 mg/ml). Diluted samples were mixed with diluted Folin-Ciocalteu and 2.5% Na₂CO₃ and the absorbance at 650 nm was measured after incubation for 1 h.

Animals and Treatments

Male C57BL/6J mice were purchased from The National Laboratory Animal Center, Thailand. The experimental protocol has been approved and all animal procedures were conducted in accordance with the guidelines approved by the Animal Ethics Committee of Khon Kaen University. Mice were divided into three groups: control ($n=6$), a low dose ($n=7$) and a high dose group ($n=5$). In the low dose and high dose groups, mice were fed with *S. gratum* aqueous extract at a dose of 0.25 or 1 g/kg/day once a day for 30 days, while the control group received only distilled water. At the end of treatment, anesthesia was induced and then the peritoneal and chest cavities were immediately opened. The liver was quickly harvested and washed in cold 100 mM phosphate buffer (pH 7.4) containing 2 mM MgCl₂. About 25–35 mg of the liver sample was used for isolation of RNA with Trizol reagent according to the manufacturer's guidelines. The remaining tissue was prepared to obtain microsome and cytosol parts by homogenizing in 4 volumes of MgCl₂-phosphate buffer on ice. After that the homogenate was centrifuged at 15,000 *g* for 30 min at 4 °C and the obtained supernatant was further centrifuged at 100,000 *g* for 1 h at 4 °C to collect the cytosol and microsome fractions.

Measurement of γ -GCL and HO-1 Activity

γ -GCL activity was determined in liver cytosol using a fluorescence method as previously described [18]. HO-1 activity was measured in the liver microsomes using spectrophotometry as described previously [19]. Briefly, the microsomes were added to a reaction mixture containing 25 μ M hemin, 2.85 mg/ml of mouse liver cytosol prepared from 100,000 *g* supernatant (serving as a source of biliverdin reductase), 2 mM glucose-6-phosphate, 0.8 U of glucose-6-phosphate dehydrogenase and 1 mM NADPH in a total volume of 500 μ l. Following 1 h incubation in the dark at 37 °C, the reaction was terminated with the addition of 500 μ l chloroform. The bilirubin generated was measured in a spectrophotometer at A₄₆₄–A₅₃₀ ($\epsilon=40$ mM⁻¹cm⁻¹). A blank reaction contained all components except the test protein.

Total Glutathione (GSH) Content

GSH level in the liver was assayed by a previously described method [18].

Determination of mRNA Expression

Semiquantitative reverse transcription-polymerase chain reaction (RT-PCR) was used for the determination of mRNA expression level. About 7 μ g of RNA was reverse-transcribed into cDNA in the reaction mixture containing 0.5 μ g of oligo(dT)₁₅ primer, 1 mM of each dNTP, 20 U of ribonuclease inhibitor, 1 μ l of ImProm-IITM reverse transcriptase (Promega, Madison, WI, USA), 5 $\times$ reaction buffer and 3 mM MgCl₂. Then 2 μ l of produced cDNA was amplified by PCR in the mixture containing 2 mM MgCl₂, 0.4 mM of each dNTP, 0.4 μ M of GCL_h (GCL heavy chain) forward (5' AGAAGGGGAGAG GACAAAC 3') and reverse (5' AGTGATGGTGCAGA GAGCCT 3') primers [20], and 0.6 μ M of HO-1 forward (5' GCCACCAAGGAGGTACACAT 3') and reverse (5' GCTTGTTGCGCTCTATCTCC 3') primers [21], and 0.2 μ M of β -actin forward (5' TGTGATGGTGGGAAT GGGTCAG 3') and reverse (5' TTTGATGTCACGCAC GATTTC 3') primers [22] and 2.5 U of Taq DNA polymerase (Promega). PCR was performed with a DNA thermal cycle as follows: initial denaturation step for 3 min at 94 °C; 30 cycles of amplification step consisting of denaturation for 1 min at 94 °C, annealing for 1 min at 61 °C and elongation for 1 min at 72 °C; and final extension step for 10 min at 72 °C. PCR products were visualized by electrophoresis on 3% agarose gel in the presence of 0.5 μ g/ml ethidium bromide. Density of bands was measured using Gel-Pro Analyzer densitometry software.

Statistical Analysis

Data were presented as mean $\pm$ S.D. The IC₅₀ values, which were determined by a non-linear regression software program, present the concentration of a sample to scavenge 50% of the DPPH radical or to inhibit the intracellular oxygen radical formation by 50%. A Student's *t*-test was used for the determination of statistical significance. A *p* value of less than 0.05 was considered significant.

Results

Free Radical Scavenging Activities of *S. gratum* Extracts

To determine the free radical scavenging activity of the plant extracts, DPPH assay was performed and results are shown in Fig. 1. Both water and ethanolic extracts of *S.*

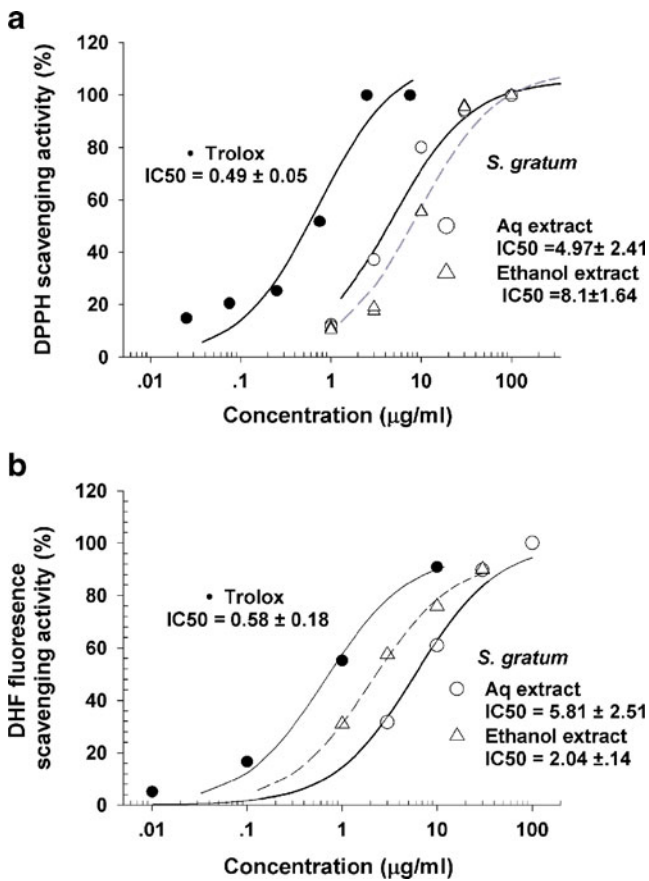


Fig. 1 Free radical scavenging activities of *S. gratum* extracts. **a** 1,1-diphenyl-2-picrylhydrazyl (DPPH) radical scavenging activity and **b** dihydrofluorescein (DHF) assay of crude extracts of *S. gratum* and Trolox as reference compound. The IC_{50} values are expressed as mean $\pm$ S.D., from 3 experiments

gratum showed strong radical scavenging activities when compared with Trolox with IC_{50} values of 4.97 ± 2.41 , 8.1 ± 1.64 and 0.49 ± 0.05 $\mu\text{g/ml}$, respectively (Fig. 1a), and aqueous extract was more potent than the ethanolic extract. Likewise, the DHF assay to determine the ability of the *S. gratum* to scavenge intracellular oxygen radical revealed a relatively high potency of both aqueous and ethanolic extracts compared with Trolox with IC_{50} of 5.81 ± 2.51 ,

2.04 ± 0.14 and 0.58 ± 0.18 $\mu\text{g/ml}$, respectively (Fig. 1b). It was noted that the ethanolic extract showed higher potency in DHF assay than the aqueous extract.

Total Antioxidant Capacity of *S. gratum* Extracts

The total antioxidant power of the extracts measured by FRAP assay and expressed as ascorbic acid equivalents was determined, and the results are shown in Table 1. The total reducing capacity measured as total phenolic content was also measured and expressed as gallic acid equivalents. The aqueous extract had relatively higher antioxidant capacity in both assays than the ethanolic extract (Table 1).

General Toxicity of *S. gratum* Extract

Both aqueous and ethanolic extracts of *S. gratum* showed potent antioxidant capacities as described above. However, the water extract was generally shown to be more potent than the ethanolic extract except for the DHF assay. Therefore the water extract was then used in the animal study. During the administration of the extract, the general appearance and behavior of the animals were monitored daily and body weight was measured weekly. The changes in body weight were comparable in the three groups of animals (data not shown). At the end of the study, mice were sacrificed and necropsy was performed. No abnormality was observed in any of the mice. In addition, the weight of major internal organs including liver, kidney, spleen and heart did not differ among the groups (data not shown), indicating that the administration of the aqueous extract of *S. gratum* was generally safe at a dose up to 1 g/kg/day.

Effect on HO-1 and γ -GCL Activity in Mice

As antioxidant and cytoprotective effects of phytochemicals may be primarily due to an induction of endogenous antioxidant enzymes, the activities of HO-1 and γ -GCL were determined in the liver of mice. Hepatic HO-1 activity was demonstrated to be significantly elevated in the high dose treatment group when compared with the control and low dose groups (Table 2). However, no significant

Table 1 The total antioxidant capacity (FRAP assay) and total phenol content of crude extract of *S. gratum*

<i>S. gratum</i> extract	Ascorbic acid equivalence ($\mu\text{g/mg}$ extract) ^a	Total phenol content (g gallic acid/100 g extract) ^b
Aqueous extract	295 ± 38	$31.3 \pm 1.0^*$
Ethanolic extract	262 ± 48	12.1 ± 0.3

^a FRAP assay was expressed as ascorbic acid equivalents per mg of the extract.

^b Gallic acid equivalents in g per 100 g of the extract.

Values are mean $\pm$ S.D. from three experiments * Significant difference at $p < 0.05$ vs. ethanolic extract.

Table 2 Heme oxygenase (HO-1) and γ -glutamylcysteine ligase (γ -GCL) activities in the liver of C57BL/6J fed with *S. gratum* aqueous extract

Enzyme activity	Control (n=6)	Low dose (n=7) 0.25 g/kg/day	High dose (n=5) 1 g/kg/day
HO-1 (pmol/mg protein/min)	11.50 $\pm$ 0.91	12.02 $\pm$ 1.85	16.50 $\pm$ 3.08*
γ -GCL (nmol/mg protein/min)	21.13 $\pm$ 4.11	24.08 $\pm$ 2.16	23.39 $\pm$ 4.36

Values are expressed as mean $\pm$ S.D.

* Significant difference at $p < 0.05$ from control and low dose group.

alteration in γ -GCL activity was observed among the three groups of mice.

Effect on HO-1 and GCL mRNA Expression in Mice

To investigate whether the effect of *S. gratum* on increased enzyme activity was due to the induction of gene expression, RT-PCR was performed. Consistent with the enzymatic activities, the expression of HO-1 mRNA tended to increase in the high dose treatment group, whereas GCL_h mRNA was not different among these animal groups as shown in Fig. 2.

Effect on Hepatic GSH in Mice

To ascertain the effect of *S. gratum* on GSH synthesis, the amount of total GSH was determined in mice liver. The results showed that the administration of *S. gratum* extract did not significantly alter the levels of hepatic GSH (0.37 $\pm$ 0.07, 0.39 $\pm$ 0.09 and 0.41 $\pm$ 0.13 μ mol/mg total protein for control, the low dose and high dose group, respectively with $p > 0.05$).

Discussion

Currently, there is growing research on the screening of medicinal plants as modern sources of new antioxidants. The present study has provided clear evidence for the strong antioxidant activities of *S. gratum* as demonstrated in the assays of free radical scavenging activity and ability to scavenge intracellular oxygen radicals. In addition to the direct antioxidant effect, the extract of *S. gratum* was able to induce the body's endogenous antioxidant and cytoprotection system.

Previous studies have indicated that the major components of plants in the genus *Syzygium* belong to the group of phenolic compounds [23]. Consistent with these reports, we found that the extracts of *S. gratum* possessed a very potent antioxidant property. The antioxidant activity of the extracts is in the same order of magnitude of the pure compounds, i.e., ascorbic acid. These data suggest that *S. gratum* is a rich source of antioxidant compounds.

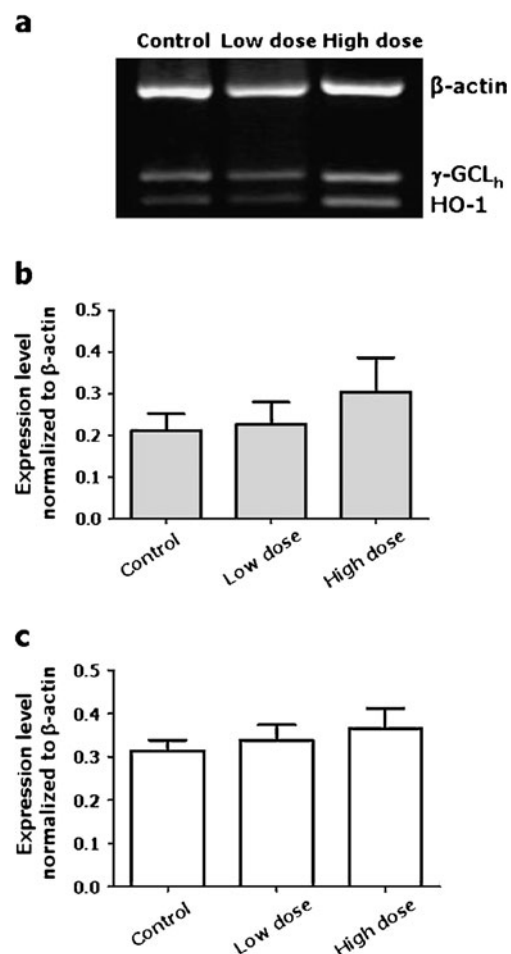


Fig. 2 Expression of hepatic heme oxygenase (HO-1) and glutamylcysteine ligase (GCL_h) mRNA. C57BL/6J mice were fed with an aqueous extract of *S. gratum*, in low dose group (0.25 g/kg/day) and high dose group (1 g/kg/day) ($n=3$ each). A control comprising three animals was fed vehicle only. Gene expression was determined by reverse-transcription chain reaction analysis and β -actin was used as an internal control. The DNA bands were quantified by densitometry software and expressed after normalization to β -actin expression. **a** Representative polymerase chain reaction products of target genes from each group were analyzed on 3% agarose gel containing ethidium bromide for visualization. **b** Expression level of the HO-1 gene normalized to β -actin. **c** Expression level of the GCL_h gene normalized to β -actin. Values are expressed as mean $\pm$ S.D.

A body of evidence suggests that diverse phytochemicals with antioxidant properties can protect against oxidative damage by directly neutralizing reactive oxidants and/or may increase the capacity of endogenous antioxidant defenses [4, 7]. Moreover, it has been reported that the antioxidant capacity of these compounds is not necessarily correlated with their activity in the induction of antioxidant systems [24]. In this study, besides directly reacting with reactive oxygen species as mentioned above, we found that *S. gratum* could significantly increase the activity of hepatic HO-1 enzyme, one of the most potent enzymes in the endogenous antioxidant defense system. Consistent with the enzyme activity, the expression of HO-1 gene tended to be elevated by treatment with the extract of *S. gratum*. Accumulating data indicate that HO-1 plays functional roles in conferring cytoprotection on cells [11]. HO-1 gene expression is strongly induced by agents or conditions that increase oxidative stress including heavy metal ions, hypoxia, heat shock, ischemia, cytokines, inflammation and others [25]. In the present study, mice treated with the aqueous extract of *S. gratum* at a dose of 1 g/kg/day did not show any signs of toxicity or abnormalities, thus supporting the direct effect of this plant on the expression of HO-1. Our result is in agreement with a previous study by Lee et al. [26] who found that the extract of *Syzygium formosanum*, a plant under the same genus as *S. gratum*, containing abundant phenolic constituents, exhibits an inductive effect on HO-1 expression. Thus, our results have clearly demonstrated that *S. gratum* exerts its antioxidant effects by up-regulating the antioxidant defensive system as well as by directly neutralizing the reactive oxidant species. The present result lends support to our previous study on the cardiovascular protection of *S. gratum* in phenylhydrazine-induced impairment of vascular function [16].

In contrast to the HO-1 enzyme, we did not observe a significant alteration in activity and expression of hepatic GCL_h in mice treated with *S. gratum* extract. Consistently, the amount of hepatic GSH was also not altered. The expression of HO-1 and GCL are usually induced by similar factors such as oxidative stress or depletion of GSH. It has been demonstrated that the transcriptional regulation of GCL_h is tissue specific and partly dependent on intracellular GSH levels [20]. Thus, it is possible that *S. gratum* may increase GCL expression in other organs, but not in the liver, and may only be induced under substantial conditions of oxidative stress.

In conclusion, the present study provides evidence supporting that *Syzygium gratum* is a potential source of natural antioxidants. The consumption of *S. gratum* was found to result in the promotion of the endogenous antioxidant defense system by inducing HO-1 activity with no toxicity observed. Furthermore, it is possible that *S.*

gratum may stimulate the expression of other defensive enzymes and its cellular and tissue protection may be achieved via the concert action of the multiple enzymes. Thus, *S. gratum* which is a local dietary vegetable can be considered as a safe food supplement with potential in prevention of oxidative-stress related disorders. Moreover, the plant might be a good candidate as daily vegetable for health promotion and might be a new natural source in developing antioxidant compounds.

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